

VERA — Vision-Enabled Reasoning Agent — System Architecture
v1.42.0 — Voice command pipeline + STT controls across CLI/Web/OpenClaw

Hardware (Raspberry Pi 5)

- reSpeaker XVF3800 (USB mic array)
- Pi Cam Module 3 Wide (CSI-0)
- TMP117 (I²C 0x48)
- NF-A6x25 PWM Fan
PWM ← GPIO13 (kernel HW PWM, 25 kHz)
TACH → GPIO6 (2 ppr)
- Hailo-8 AI HAT+ (26 TOPS, PCIe)
- DS3218 Servo (PCA9685 ch15)
- reSpeaker XVF3800 output → attached speaker

Drivers (src/)

- audio/input.py (USB)
- vision/camera.py (picamera2) (CSI)
- thermal/tmp117.py (I²C)
- thermal/fan_tach.py (I²C edge cb)
- thermal/fan.py (sysfs HW PWM + I²C fallback)
- perception/hailo_inference.py (VDevice wrapper) (PCIe)
- motion/servo_controller.py (I²C/PWM)
- audio/version_annotator.py (USB)
- thermal/thermal_manager.py (receives from thermal/tmp117.py, thermal/fan_tach.py, thermal/fan.py)
- perception/object_detector.py (YOLOv8s + letterbox cache) (receives from thermal/fan.py, perception/hailo_inference.py)
- perception/face_detector.py (SCRFD decode + Haar fallback) (receives from thermal/fan.py, perception/hailo_inference.py)
- audio/tts.py (espeak-ng) (receives from audio/version_annotator.py)
- audio/output.py (receives from audio/tts.py)

External processes

- ~/local/share/desktop-assistant/telemetry.db (SQLite ring) faces.db (identity registry) thumbs/ (64x64 JPEG face crops) → writes faces.db
- scripts/desktop-assistant (CLI: ping / status / pan / say / watch / ...) → writes telemetry.db
- 3rd-party subscribers (dashboards, future ML pipelines) → ZMQ REQ/SUB
- Web Browser http://<pi>:8080 → ZMQ SUB

